

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS PRODUCT DEPARTMENT
SALEM, VA. 24153

INSTRUCTIONS

GEK-40676

MKII MARINE

TURBINE CONTROL

SUPPLEMENT

GENERAL ELECTRIC

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Reference Drawings:

245A2260	Sh.	#1	GEK-40676	Sh.	#5
"	"	2	"	"	6
"	"	3	"	"	7
"	"	4	"	"	10
"	"	5	"	"	11
"	"	6	"	"	16
"	"	7	"	"	17
"	"	8	"	"	18
"	"	9	"	"	19
"	"	10	"	"	20

Abbreviations

A	- And
ACOM	- Analog Common
ADJ	- Adjustment
CCW	- Counter-Clockwise
COM	- Common
COMP	- Comparator
CW	- Clockwise
DR	- Detector
EXT	- External
FF	- Flip Flop
IC	- Integrated Circuit
L	- Logic
LIQ	- Liquid
LNG	- Liquefied Natural Gas
MAN	- Manuvering
MIN	- Minimum
N	- NOR
PRESS	- Pressure
PROPUL	- Propulsion
R	- Resistor
REF	- Reference
RPM	- Revolutions Per Minute
SCOM	- Special Common
STAB	- Stability
SW	- Switch
TP	- Test Point

CARD PIN NOMENCLATURE

(The following list will be of assistance in "reading" the card symbols. The abbreviations used are intended as cryptic notations and are not necessarily rigorous).

AG and $\overline{AG}$	- Date Driver Signal for Ahead
AHD	- Logic Signal for Ahead
ALLG	- All Gas Signal
ALLL	- All Liquid Signal
AT SEA	- Logic Signal for Automatic Control Mode
AUTL and $\overline{AUTL}$	- Logic Signals for Auto Transfer to Liquid
BG and $\overline{BG}$	- Gate Signal for Bridge Power and Pitch Control
BLO and $\overline{BLO}$	- Blow Off Valve Control Logic
BR1	- Logic Signal to Switch PLUB to Buffer Amplifier
BUFF	- Buffer Amplifier Input
COFF and $\overline{COFF}$	- Heating Value Compensator Off
CON and $\overline{CON}$	- Heating Value Compensator On
DN and $\overline{DN}$	- Digital Setpoint Down
DS	- Digital Setpoint Input
FF	- Fuel Flow Feedback Signal
FRAT	- Fuel Ratio Output
FRATJ	- Fuel Ratio Input
GAR	- Gas Available or Requested
GASA	- Gas Available
GASR	- Gas Requested
GASZ and $\overline{GASZ}$	- Gas Zero
GFC	- Gas Fuel Command
GFCB	- Gas Fuel Purge Bias
GFCM	- Gas Meter Output
GLID	- Gate Signal for Low Idle

GMA and $\overline{\text{GMA}}$	- Gas Minimum Available
GMAN	- Gate Signal for $\overline{\text{UFM}}$ and Manuver Mode
$\overline{\text{GMD}}$	- Inverse signal for Gas Minimum Detected
GMR	- Gas Minimum Request
GND	- Ground
GSEA	- Gate Signal for $\overline{\text{UFS}}$ and Sea Mode
IGA	- Inhibit Gas Available
IGP	- Inhibit for Gas Purge
ILP	- Inhibit for Liquid Purge
INHL	- Logic Inhibit for Torque Limiting
INV	- Inverse
LFC	- Liquid Fuel Command
$\overline{\text{LFCB}}$	- Inverse Liquid Fuel Purge Bias Signal
LFCM	- Liquid Meter Output
LID	- Logic Signal Switching Feedback from FF to NHP
LIQZ and $\overline{\text{LIQZ}}$	- Liquid Zero
LO	- Liquid Lockout
LVP	- Pitch Control Signal
M	- Reference Level Terminal
MAN	- Logic Signal for Manuvering Mode
MIN	- Minimum VCE Signal
MPA & MPB	- Calibration Inputs to Multiplier
MSQ	- Calibration Input to Multiplier
NHP	- High Pressure Turbine Speed Signal

NLP	- Low Pressure Turbine Speed Signal
NORM and INV	- Normal and Inverse Signals to Comparator
NTFC	- Negative Total Fuel Command
PC(10)	- Heating Value Compensator Input
PC(46) and $\overline{PC}$	- Power Control Mode Logic Signals
PITCH	- Pitch Reference Signal
PIV	- Inter Value Pressure
PLP	- Propulsion Lever Signal
PLV	- Propulsion Lever Reference Signal
PLVB	- Propulsion Lever Signal from Bridge
PLVE	- Propulsion Lever Signal from Engine Room
PPC	- Power and Pitch Programming Signal
$\overline{PR}$	- Power Reset
$\overline{PROD}$	- Product Signal
PWR	- Power Programming Input
REF	- Reference
SEA	- Input to Switch Pitch Control to Auto
SEC	- Second
SG and OSG	- Gate Signal for at Sea Mode
SJ	- Summing Junction
SJA, B, C	- Adjustable Pitch Servo Bias
SJ1	- Input from Heating Value Compensator
SP	- Set Point Input
SQO	- Square [illegible] Multiplier Output
$\overline{STOP}$	- Input [illegible] to Auto
$\overline{TDGP}$	- Inverse Delayed [illegible] Signal
$\overline{TDLP}$	- Inverse Delayed Liquid Purge Signal

TFC	- Total Fuel Command Signal
TJ	- Test Jack
$\overline{\text{TRG}}$	- Inverse Transfer Enable Gas
TRGC	- Transfer to Gas Command
$\overline{\text{TRL}}$	- Inverse Transfer Enable Liquid
TRLC	- Transfer to Liquid Command
TT	- Transfer Timing
TTC	- Transfer Timing Clamp
$\overline{\text{UFM}}$	- Logic Signal for Untreated Fuel Inhibit (Manuver)
$\overline{\text{UFS}}$	- Logic Signal for Untreated Fuel Inhibit (At Sea)
UP	- Up Digital Setpoint
VCE	- Fuel Flow Reference Signal
WG	- Actual Gas Flow
WL	- Actual Liquid Flow
WT	- Actual Total Fuel

POWER PROGRAMMING

Power programming is provided by the SPPA and SPPB cards

SPPA CARD

The SPPA card generates a power signal for marine gas turbines based on the position of the propulsion control lever. This function is performed by first generating an absolute value for the set reference. The cube of this set reference is generated to give a power reference voltage. The power reference signal is then amplified in the specific slopes and intercepts for the "At Sea" and the "maneuvering modes". The modes are determined by the "sea" and a "man" gate. In the final amplifier feedback there is a torque limiting function which limits the power signal when the low pressure speed is low.

The absolute value circuit accepts a set reference signal from the propulsion control lever. This reference signal is positive for ahead positions and astern positions. In the ahead mode, the output of the absolute value circuit varies in an analog manner with the input to produce a positive output. In astern operation, the output varies to a preset maximum astern value, which again is a positive voltage out of the absolute value circuit.

The cubic function generator creates an output signal which is proportional to the cube of the input. (Propulsion lever position) The scale factor of the circuit is adjustable so that different voltages may serve as unity reference. The output of the cubic function generator is gated through one of two resistor scaling networks depending on the mode of operation; "at sea" or "maneuvering". On the SPPA card are two logic gates; the "At Sea" gate and the "Maneuvering" gate. When the At Sea gate is enabled then the output of the cubic function generator is gated through a resistor network which establishes the slope and intercept (this intercept adds to maneuver intercept on SPPB) for the power reference signal when in the "at sea" mode of operation. When the Maneuvering gate is enabled, the slope only for the power reference is established.

The intercept for maneuver mode is located on the SPPB card. An untreated fuel command input "OUFS" and "OUFM" is available on this card to limit the power level available when running with untreated fuel.

After the signal goes through the gated resistor network, it is amplified to generate the power programming output signal.

The power programming signal is also a function of the speed of the low pressure set shaft which performs a torque limiting function. If NLP is low, the torque limit reduces the Power Programming signal. The torque limit point (offset) and GAIN of this function is also adjustable on this card. This limits the called for power so that rated torque is not exceeded at any speed.

SPPB CARD

The SPPB card generates the fuel control signal (VCE) based on the Power Programming or command signal and a feedback signal.

In normal operation the feedback signal is based on fuel flow. In the low idle mode a logic 1 is on the LID input, junction field effect transistors (JFET's) close the gate disabling the fuel feedback and gates through the high pressure speed signal, NHP, for the feedback signal. This will regulate NHP to a preset value of speed, which is adjustable on this card. The adjustment for the maneuver mode intercept is also located on this card

Fuel flow is controlled by the VCE signal. The power control loop regulates VCE through a min select circuit as a function of the power reference signal and the feedback signal. This VCE power control amplifier has selectable compensation as well as a gain adjustment of 10 to 1. When the power amplifier is controlling VCE, it turns on a comparator amplifier which gives logic level outputs to indicate power control of turbine. This power control amplifier also receives a minimum input from the start-up card which prevents the power amplifier from reducing VCE below a preset minimum.

Also on this card is a Blow Off Amplifier which compares the current required to hold VCE at a minimum to an adjustable reference. When Minimum current exceeds the reference, a trip condition is indicated by a logic 1 on the BLO signal.

SPPC (FUEL VARIATION) CARD

The SPPC card provides an analog signal which modifies the power reference for variation of fuel BTU rating. Without fuel compensation, variations in in boil-off gas nitrogen context varies the BTU rating or heating value of the gas fuel. This results in unwanted speed variation. The SPPC card, when enabled, provides an adjusting voltage which offsets these fuel flucuations.

To perform the compensating function, the SPPC card has the following circuits:

1. Logic gating which enables, disables the compensating circuit and the digital servo reference.
2. Fuel compensation integrator which generates an analog signal based on the differential between a reference signal and the NLP signal (for the "maneuvering" mode) or the PITCH signal (for the "at sea" mode).
3. Propulsion lever motion detector which detects any change in the propulsion lever and turns the compensating function off if any such motion occurs.

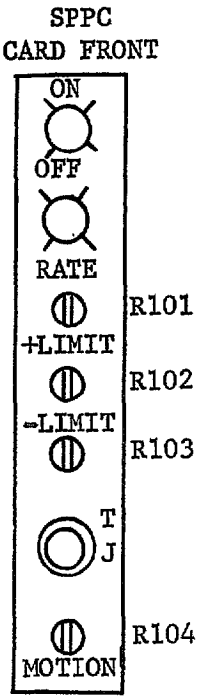
When fuel variation compensation is off, the off light on the card front is illuminated. Also, the set point is allowed to track the NLP or PITCH reference signal. This causes the compensation signal, OUT, to be at zero volts. When fuel compensation is selected (by depressing a momentary pushbutton), the on light illuminates, the set point is disabled from following a reference and remains at the NLP speed at the time compensation was selected and the fuel compensator integrating amplifier output (OUT) is enabled.

Assume operation in the Maneuvering mode and that fuel compensation is <u>not</u> selected. A voltage based on NLP (PITCH in the "At Sea" mode) is used as a reference for the set point control. The set point is equal in magnitude but opposite in polarity to the reference while it is tracking or following the reference. The input to the compensating amplifier is the sum of the set point

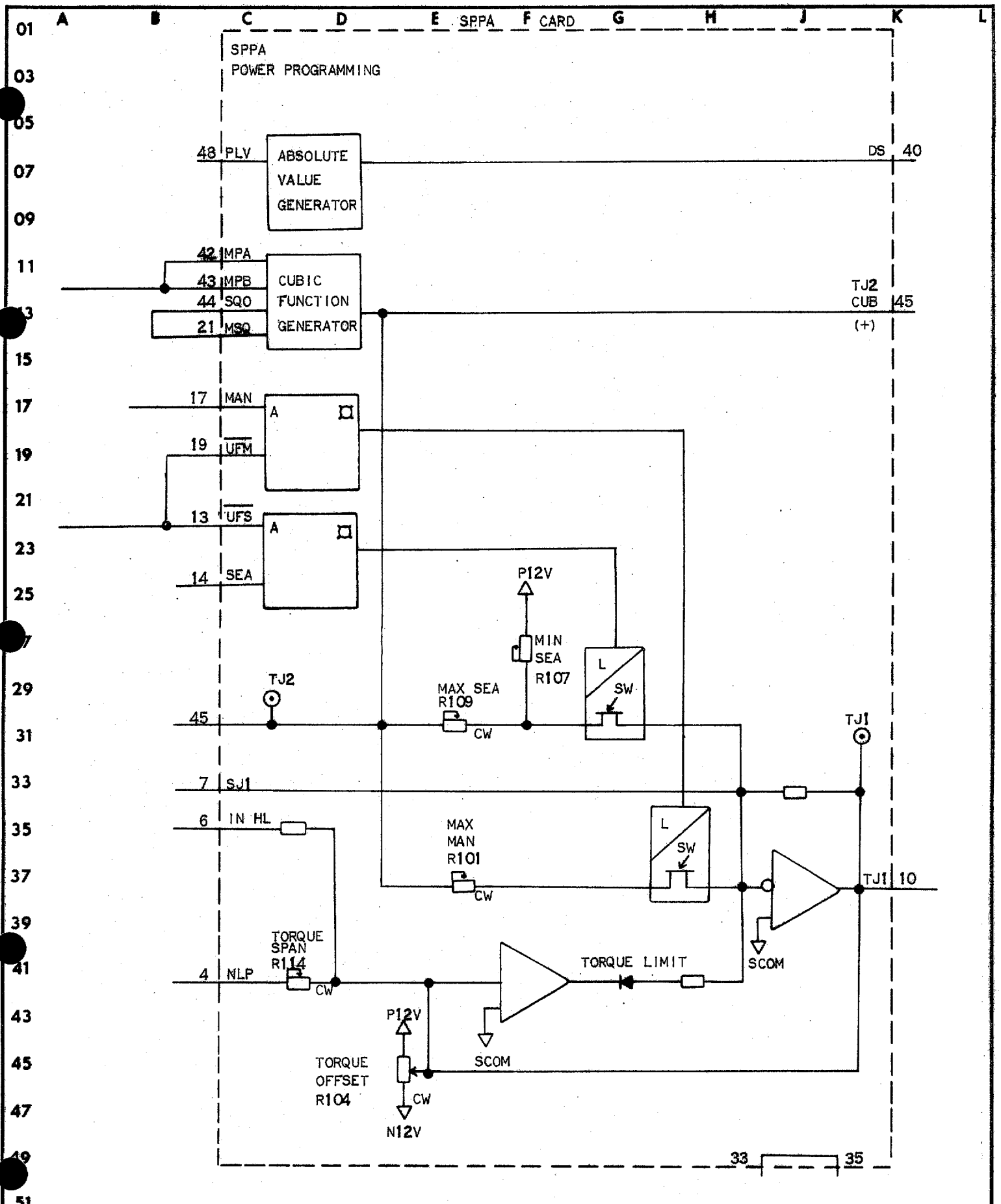
and the reference. This sum is zero. Also when compensation is turned off an FET switch places a very large feedback resistor in the feed back of the amplifier to cause the output integrator to rapidly return to zero so that compensation can again be selected.

Now assume that compensation is selected. The set point immediately stops following the reference of NLP (or PITCH). The resistor is removed from the amplifying circuit feedback path. If the reference varies from its value at the time that compensation was selected, then an error signal is generated at the input to the compensating integrator amplifier. The integration rate along with compensation limits are adjustable on the output amplifier on the SPPC card. This signal may be used to modify the power reference signal in either the positive or negative direction, thus increasing or decreasing fuel flow via VCE.

Any movement of the propulsion lever is detected by a circuit on this card. If the propulsion lever is moved a logic signal is generated by the propulsion level motion detector. The logic signal de-selects fuel compensation if it was previously selected. If the compensation function was not previously selected, then the logic signal performs no function. The sensitivity of this detector is adjustable on this card.



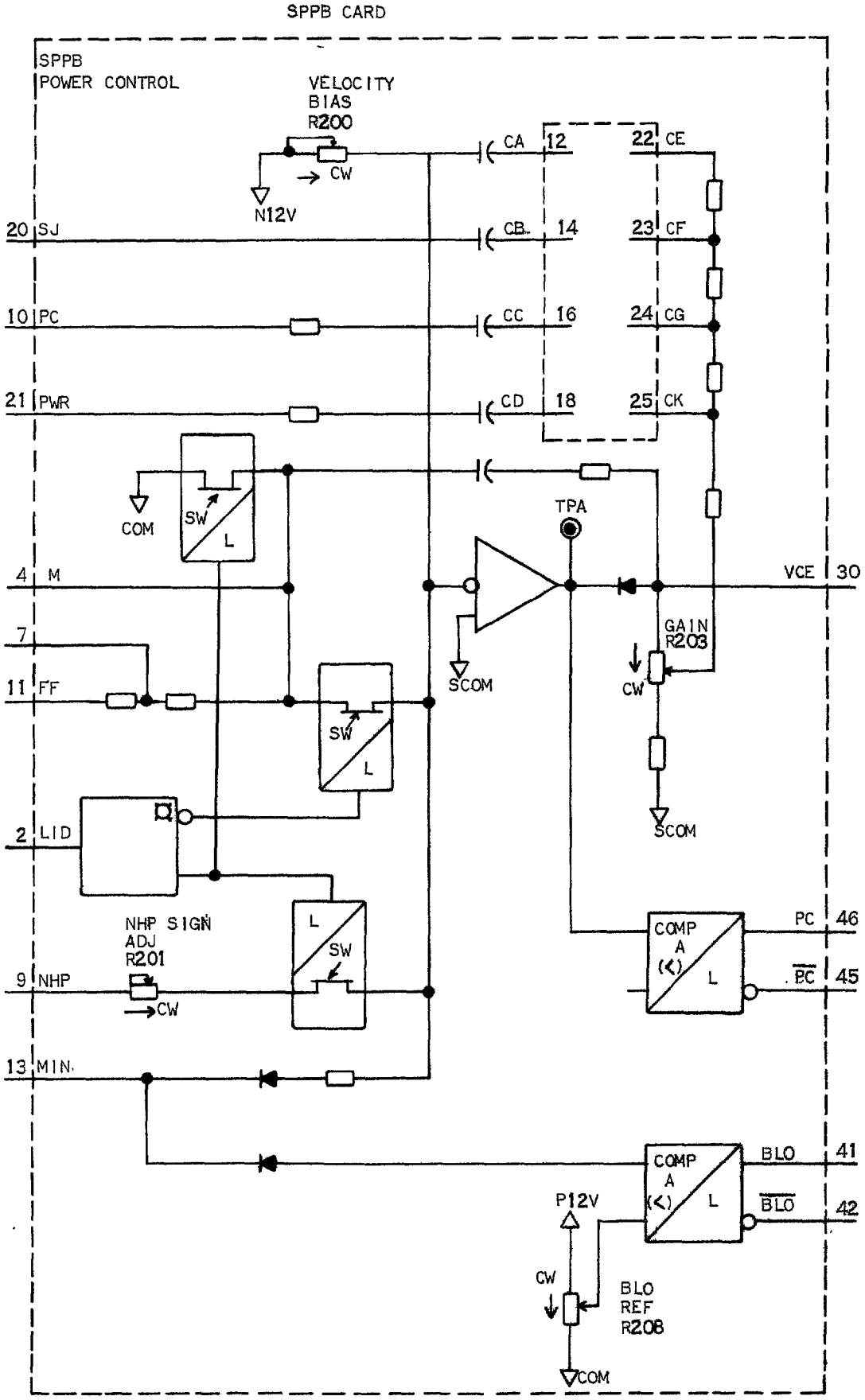
FORM 1C 2390 2-72



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY GEK-40676, SH. 45

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS DRR	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM
REV. 2	ISSUED 6-3-74			FIRST MADE FOR GEK-40676		245A2260
REV. 3	MADE BY C. Andrews P.A. Comisco			I.C. NO.		CONT. ON SH. 2 SH. NO. 1

FORM 1C 2390 2-72



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY GEK-40676, SH. #6

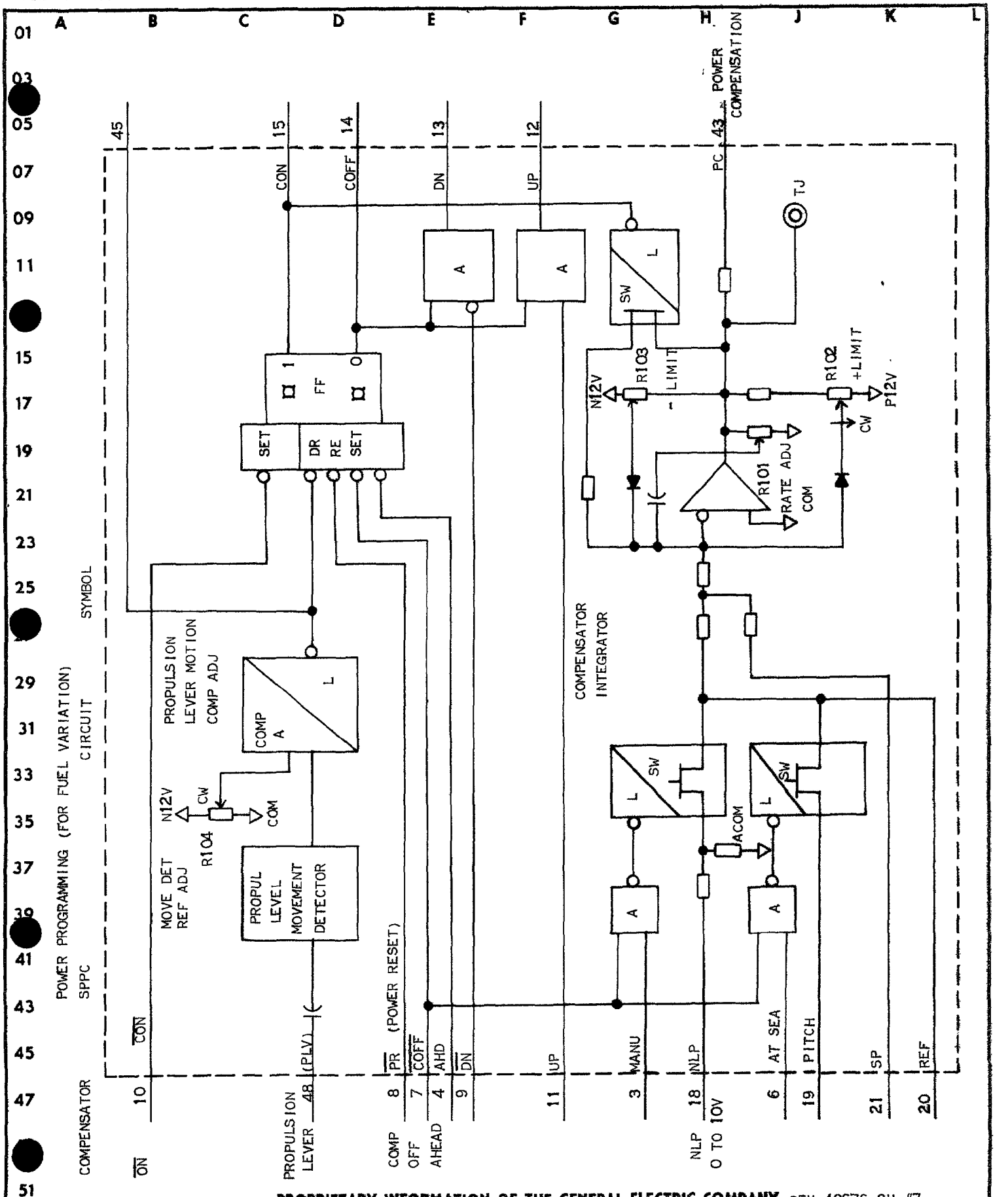
REV. 1	REV. 4	REV 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED 6.3.74			FIRST MADE FOR GEK-40676	
REV. 3	MADE BY			I.C. NO.	

ELEMENTARY DIAGRAM

2 4 5 A 2 2 6 0

CONT. ON SH. 3 SH. NO. 2

FORM 1C 2390 2-72



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY GEK-40676, SH. #7

REV. 1	REV. 4	REV 5	PRINTS TO	APPROVALS
REV. 2	ISSUED 6-3-74			FIRST MADE FOR GEK-40676
REV 3	MADE BY			I.C. NO.

GENERAL ELECTRIC
INDUSTRY CONTROL DEPT.
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM

245A2260

CONT. ON SH. 4 SH. NO. 3

PITCH CONTROL (SPKA CARD)

This card is used to control the propeller pitch position for the marine gas turbine. It has circuitry to perform the following functions;

1. Connect the propulsion control signals from the bridge or the engine room thru a switch and buffer to the power control and pitch control circuits.

2. Generate the pitch signal in the "At Sea" operating mode. In this mode the propeller pitch is automatically modulated to hold rated shaft RPM for any position of the propulsion lever.

3. Generate the pitch signal in the "Maneuvering" mode. In this mode the propeller pitch is varied in response to the propulsion lever position.

Propulsion Control Buffer

The propulsion control voltage can be applied in two modes (Fig. 1). Input BUFF provides a signal from the bridge propulsion lever (0 to $\pm$ 8V) which directly controls both power programming on the SPPA1 card and pitch programming on this card. Output PPC is a positive voltage for the ahead operation and a negative voltage for the astern operation (0 to $\pm$ 8V).

In the second mode of operation the PPC output is controlled by either the bridge or the engine room. A "1" on input BR1 will provide for bridge control and allow input PLVB (0 to $\pm$ 8V) to be applied to the buffer amplifier. A "0" on input BR1 will switch control to the engine room and input PLVE (0 to $\pm$ 8V) will be applied to the buffer amplifier. A simplified diagram follows.

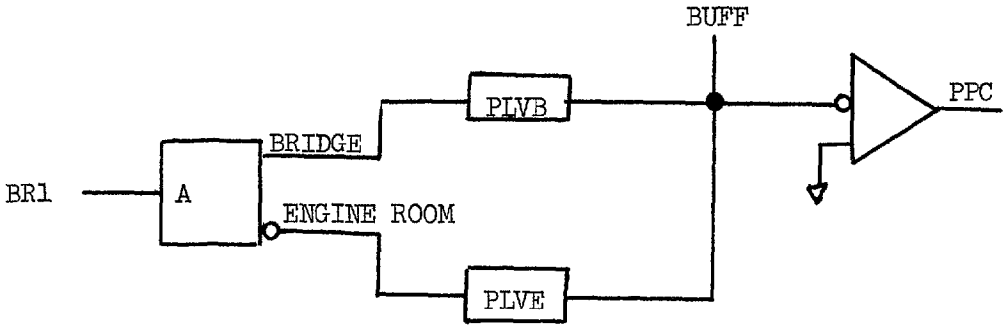


FIG. 1

At Sea Mode

In the at sea mode the propeller pitch is automatically modulated to hold rated shaft RPM. A logic "1" on the sea and stop inputs will enable the at sea mode (Fig. 2). A logic "1" on the AHD input will provide for ahead operation and output PITCH will be a positive voltage (0 to +6V). In astern operation input AHD will be a logic "0" and output PITCH will be a negative voltage (0 to -6V).

Maneuvering Mode

In the maneuvering mode the propeller pitch is varied in response to the propulsion lever position signal. A logic "0" on the sea input will enable the maneuvering mode (Fig. 2). The signal applied to input LVP is a bipolar signal, a positive voltage at LVP will provide for ahead operation and output PITCH will be a positive voltage (0 to +6V). A negative voltage at LVP will enable astern operation and output PITCH will be a negative voltage (0 to -6V).

FORM IC 2390 2-72

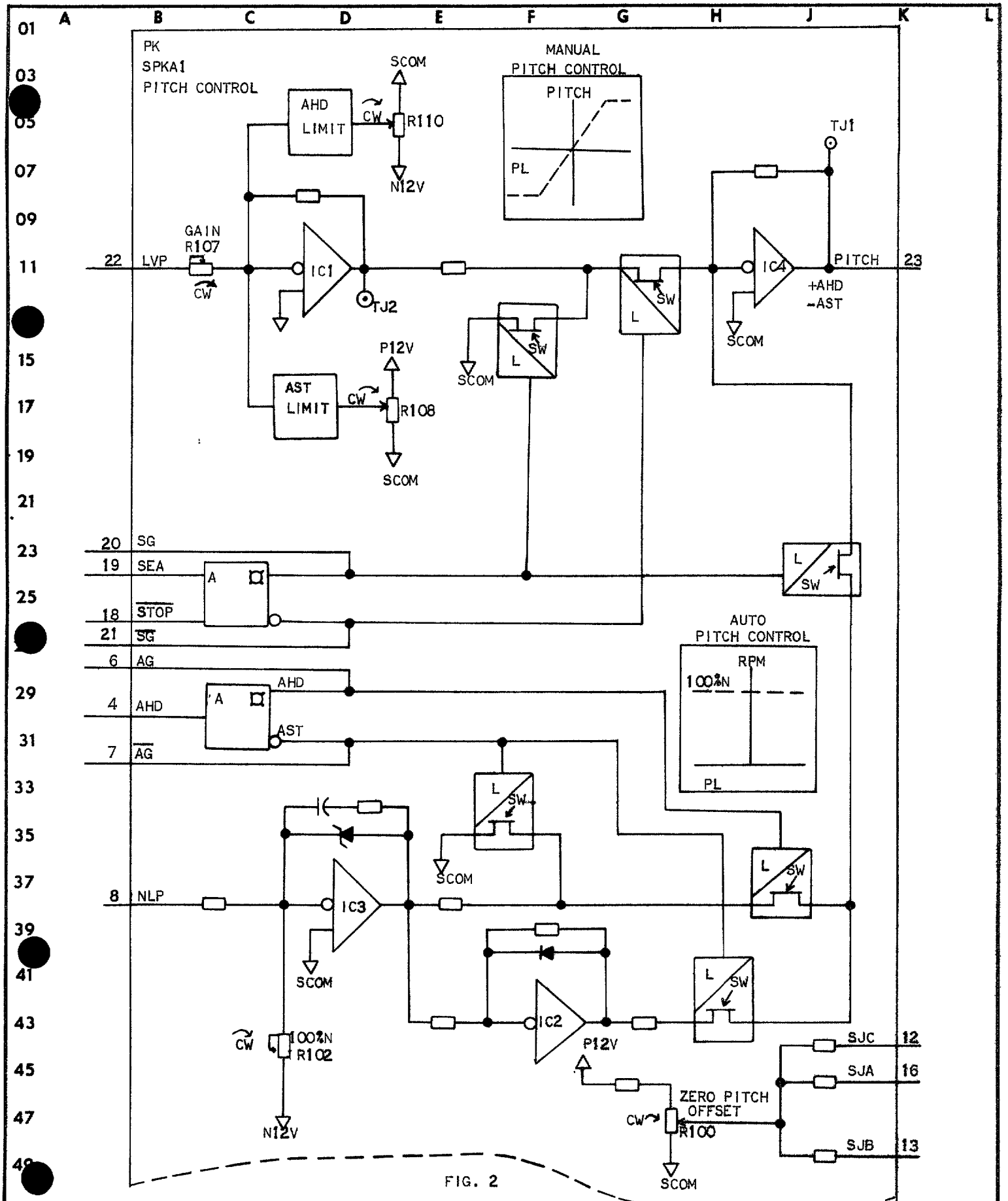


FIG. 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY GEK-40676, SH. #10

REV. 1	REV 4	REV. 5	PRINTS TO	APPROVALS DRR	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV 2	ISSUED 6-3-74			FIRST MADE FOR GEK-40676	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	245A2260
REV. 3	MADE BY G. Andrews P. Norman			I.C. NO.		CONT. ON SH 5 SH. NO. 4

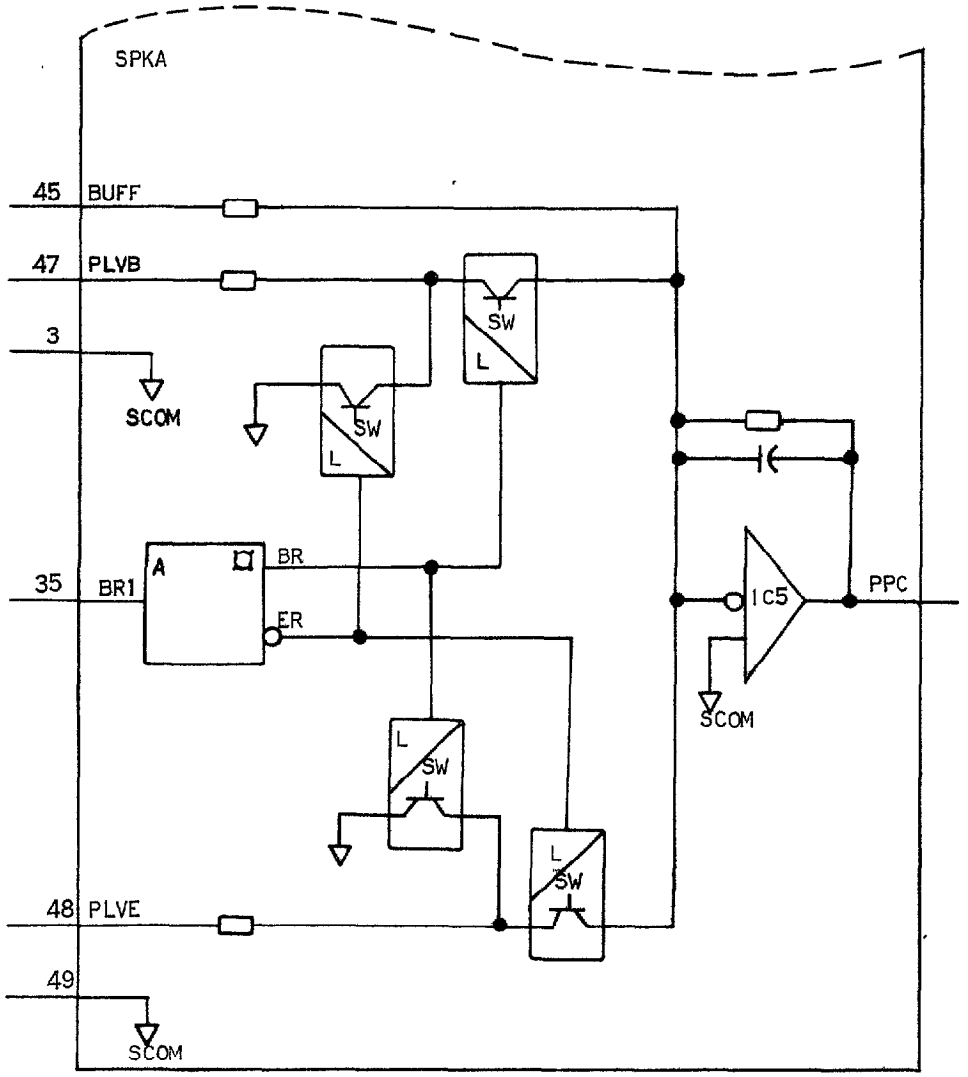


FIG. 2A
(CONTINUATION OF FIG. 2)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY GEK-40676,SH.#11

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS DKR	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM
REV 2	ISSUED 6 3 74			FIRST MADE FOR GEK-40676		2 4 5 A 2 2 6 0
REV 3	MADE BY			I.C. NO.		CONT ON SH 6 SH. NO. 5

FUEL SPLITTER (LNG TANKERS)

A fuel splitter is provided in marine applications where the gas supply is a function of the rate of gas "BOIL-OFF" in a liquefied natural gas (LNG) tanker. In order to satisfy the total fuel requirements of the turbine, the fuel splitter will continuously modulate the fuel split between the liquid and gas fuel as a function of the availability of the gas fuel. The fuel splitter must also allow for single operation on either fuel if so selected. The control functions for fuel splitting have been divided between these cards, SFKE, SFKF, and SFKG.

SFKE Card (Fig. 4)

This card contains the control logic for operating modes and purge controls. It also contains the generating circuit for the total fuel command signal (TFC).

There are four modes of operation:

1. All liquid mode
2. All gas mode
3. Dual fuel mode, using all available gas
4. Fixed ratio of gas mode, jumper selectable

To transfer from all gas mode to all liquid mode, a logic "1" is applied to ALLL and a logic "0" to ALLG. As a result TRG is a "1" starting an external liquid purge delay timer (20-90 sec.). Upon purge complete TDLP is "0" enabling transfer to liquid (TRLC). When transfer is complete gas zero (GASZ) becomes "1" which sets the gas purge flip flop inhibiting transfer to that fuel.

In transfering from mixed fuel to all liquid, the liquid purge flip flop is reset and GAR is controling the amount of each fuel. An immediate transfer to liquid is made when a logic "1" is applied to GMR, GMA, ALLL, or AUTL.

To transfer from all liquid to all gas a "1" is applied to ALLG and a "0" to ALLL. As a result $\overline{\text{TRL}}$ is a "1" starting an external gas purge delay timer (20-90 sec.). Upon purge complete TDGP is "0" enabling transfer to gas (TRGC). When transfer is complete liquid zero (LIQZ) becomes "1" which sets the liquid purge flip flop inhibiting transfer to liquid.

Transfer from mixed fuel to all gas is similar to transfer from all liquid to all gas, only the starting point is different and there is no time delay.

The total fuel command (TFC) is the product of the fuel reference (VCE) and the speed signal (NHP) when $\overline{PROD}$ is a "0". TFC equals fuel reference alone when $\overline{PROD}$ is a "1". A comparator is available on this card to enable several modes of operation.

SFKF Card (Fig. 5)

This card contains the transfer timing command circuit (TT) and generates the gas requested (GASR), gas available (GASA) signals as well as the lesser of the two (GAR). It also generates the logic signals, minimum gas requested (GMR), and minimum gas available (GMA). The gas minimum available signal is sent to an external timer to prevent using gas until pressure has returned for 15 minutes.

Boil off gas inner valve pressure (P1V) will build up until it reaches a desired pressure set on the pressure reference adjust pot. Slightly above this pressure the gas available integrator integrates positive indicating there is gas available (GASA). When the gas is used the regulator automatically assumes a level of use that keeps pressure constant at the adjustable setting.

In dual fuel mode GASR is total fuel command minus the minumum desired liquid level (5 - 20% max). This is accomplished by summing NTFC and the liquid minimum adjust pot. In the all gas mode GASR=TFC because TT rises enough to just cancel the effect of the liquid min. adj. pot.

The GAR signal is obtained through a minimum value gate of GASA and GASR.

Two comparators detect minimum gas available (GMA) or minimum gas request (GMR). Both signals are used to initiate transfer to the all liquid mode (if desired) and are adjusted by the gas minimum adjust pot (8-40%).

The transfer timing (TT) signal determines the operating mode.

1. Zero for dual fuel or fixed ratio

2. Negative for all liquid

3. Positive for all gas

For all liquid (TRLC) = "1" and TRGC = "0". This causes the gas switch to close shunting bypassing the negative current to ground. Because the liquid switch is open, a positive current is supplied an integrator which provides a constant rate of change on TT which is factory set for a specific traverse time.

FORM 1C 2390 2-72

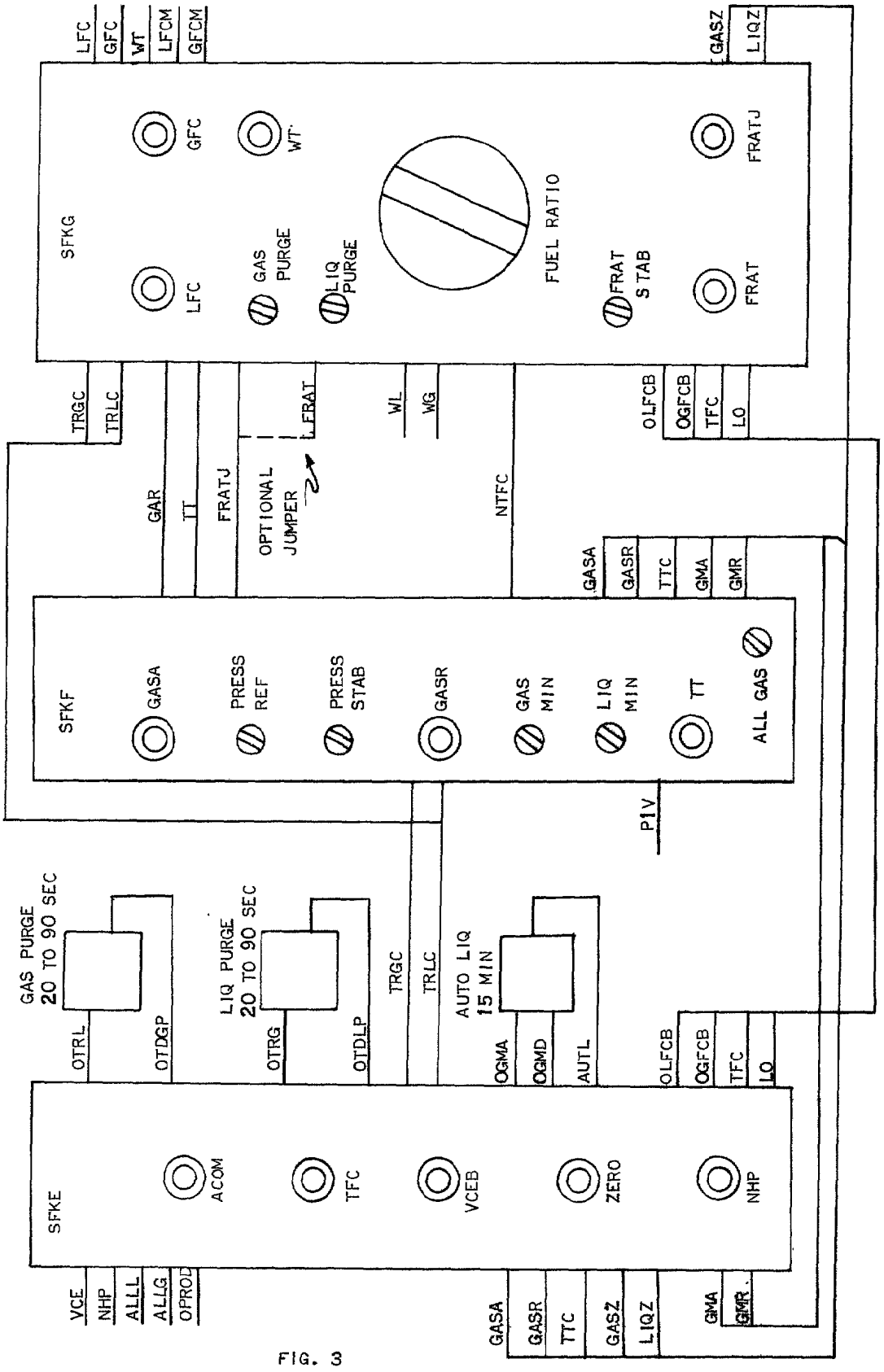


FIG. 3

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY GEK-40676, SH. #17

REV 1	REV. 4	REV 5	PRINTS TO	APPROVALS DRR	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED 6 3 24			FIRST MADE FOR GEK-40676	ELEMENTARY DIAGRAM
REV 3	MADE BY			I.C. NO.	2 4 5 A 2 2 6 0 CONT. ON SH. 8 SH. NO. 7

SFKG Card (Fig. 6)

This card takes the total fuel command (TFC) signal and splits it into a liquid fuel command (LFC) and gas fuel command (GFC) in the manner dictated by the other fuel splitter cards. Other circuits on this card are the fuel ratio regulator (FRAT), purge bias controls and the zero fuel signals for gas and liquid. Also actual liquid and gas signals are summed to provide a total actual fuel feedback signal.

In the all liquid mode (TT) is negative and will overcome any gas available or requested signal (GAR). GFC will then be driven negative insuring full shut off of the gas valve. With the liquid purge switch open LFC exactly follows TFC. Thus the total fuel command becomes a liquid full command with the gas valve fully closed.

In the mixed fuel mode TT is zero and no longer cancels GAR. With the gas purge switch open GFC exactly follows GAR. Note that gas subtracts from total fuel and thus LFC= TFC-GFC.

In the all gas mode TT is positive and through a diode applies a small bias to drive LFC negative to insure that the liquid valve is closed. On the SFKE card, GAR is made to equal TFC.

In the fuel ratio mode TT is zero and FRAT is jumpered to FRATJ which is connected to the SFKF card so that it controls GAR. An integrator is used to swing the request part of GAR over its complete range to satisfy its input requirements. The reference to the integrator is some percentage of TFC as set on the fuel ratio adjust pot. The feedback is actual gas flow (WG) from a valve position sensor. The feedback and reference must equal or the integrator moves up or down to change gas flow until they are equal.

When the liquid purge switch is closed for the purge interval, a signal controlled by the liquid purge adjust pot (0-10% fuel) is applied to the input of the liquid fuel amplifier which is not affected by the total fuel minus gas fuel amplifier output. After the purge time delay the total fuel minus gas fuel amplifier increases positive from negative. When it reaches zero, the liquid zero signal (LIQZ) becomes a zero and through logic on the SFKE card the liquid purge switch will open. Thus liquid will drop back to zero and now the transfer proceeds normally. The gas surge circuits are the same. A FRAT STAB adjust pot on the card front controls the rate on integration for the FRAT signal.

TYPICAL TRANSFERS ARE SHOWN IN FIGURE 1 AND 2 BELOW.

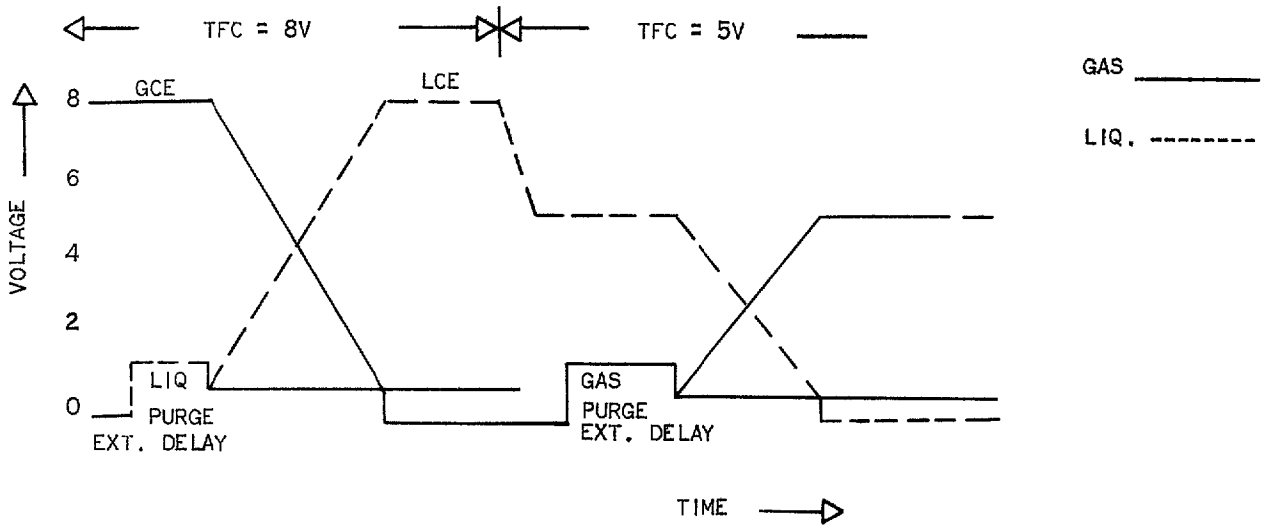


FIGURE 1 TYPICAL FUEL TRANSFER (ALL GAS OR ALL LIQUID MODES)

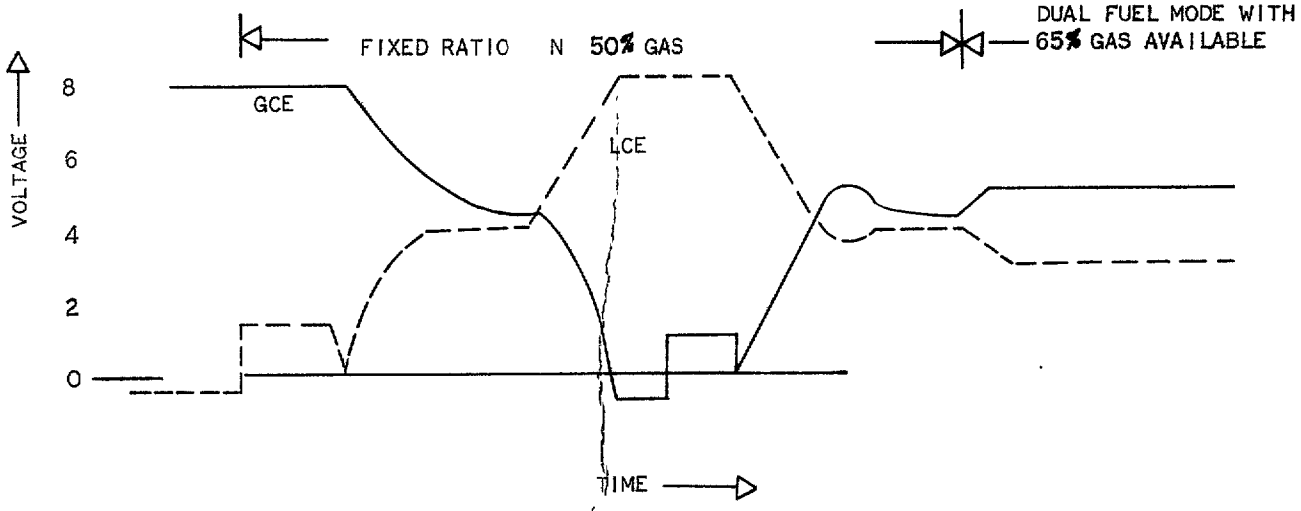


FIGURE 2 TYPICAL FUEL TRANSFER (FIXED RATIO OR DUAL FUEL MODES)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY GEK-40676, SH. #16

REV. 1	REV 4	REV. 5	PRINTS TO	APPROVALS DKR	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM
REV. 2	ISSUED 6-3-74			FIRST MADE FOR GEK-40676		245A2260
REV. 3	MADE BY C. Andrews			I.C. NO		CONT ON SH. 7 SH. NO. 6

The all gas mode is similar to all liquid but with opposite polarities. TT goes to the positive clamp which must be adjusted to cancel liquid minimum setting.

With both switches closed (TRLC=0 and TRGC=0) there is no input to the integrator so TT drives to zero at approximately the same rate as above.

Diode clamps are provided to limit TT voltage in the various modes.

FORM 1C 2390 2-72

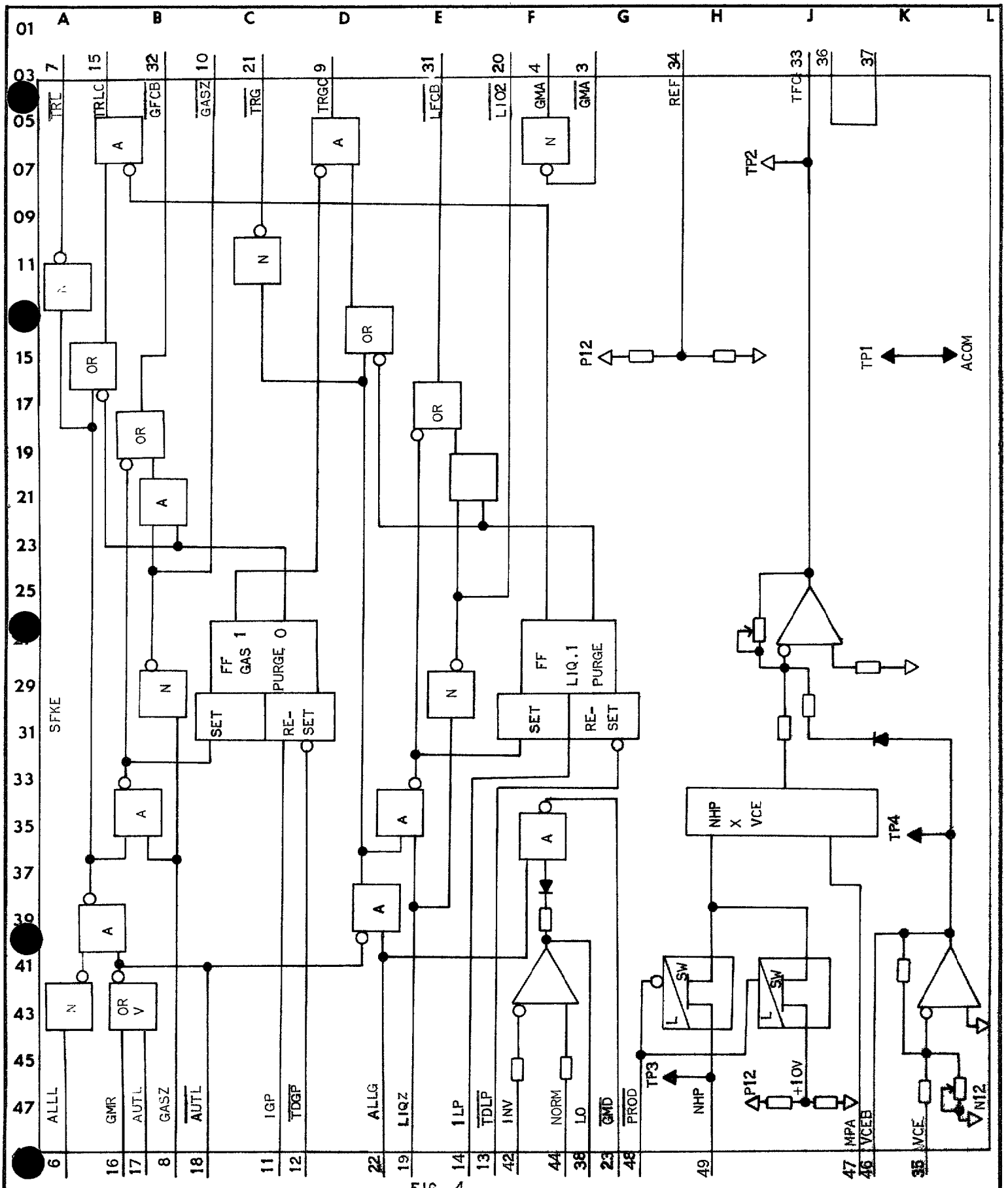


FIG. 4

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY GEK-40676, SH. #18

REV 1	REV 4	REV 5	PRINTS TO	APPROVALS DRR	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV 2	ISSUED 6-3-74			FIRST MADE FOR GEK-40676	ELEMENTARY DIAGRAM 245A2260 CONT. ON SH. 9 SH. NO. 8
REV 3	MADE BY C. Andrews			I.C NO	

FORM 1C 2390 2-72

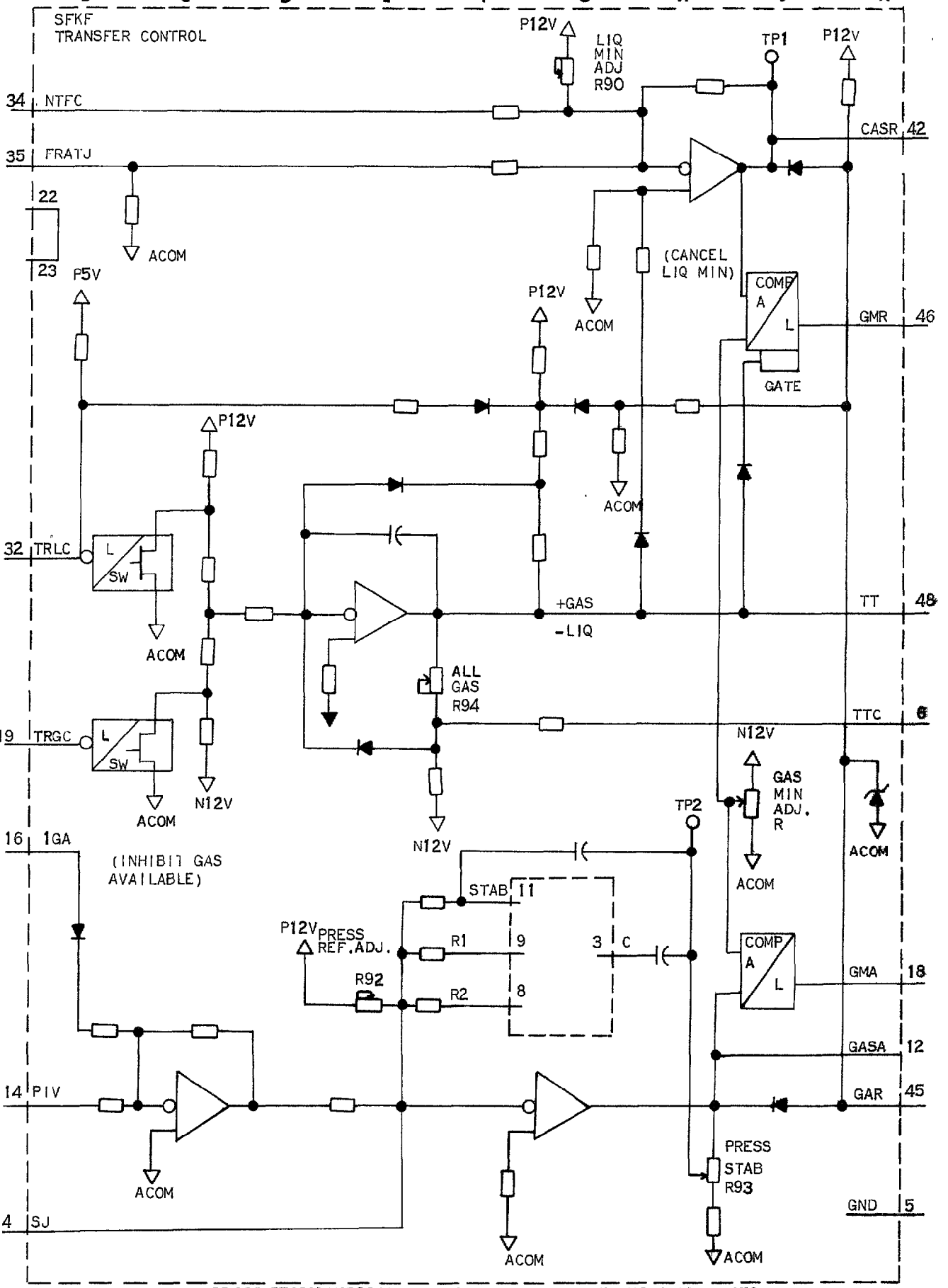


FIG. 5

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY GEK-40676, SH. #19

REV 1	REV 4	REV. 5	PRINTS TO	APPROVALS JRR	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV 2	ISSUED 6 3 74			FIRST MADE FOR GEK-40676	
REV 3	MADE BY S. Andrews P. Norris			I.C. NO	

ELEMENTARY DIAGRAM

245A2260

CONT ON SH 10 SH NO 9

FORM 1C 2390 2-72

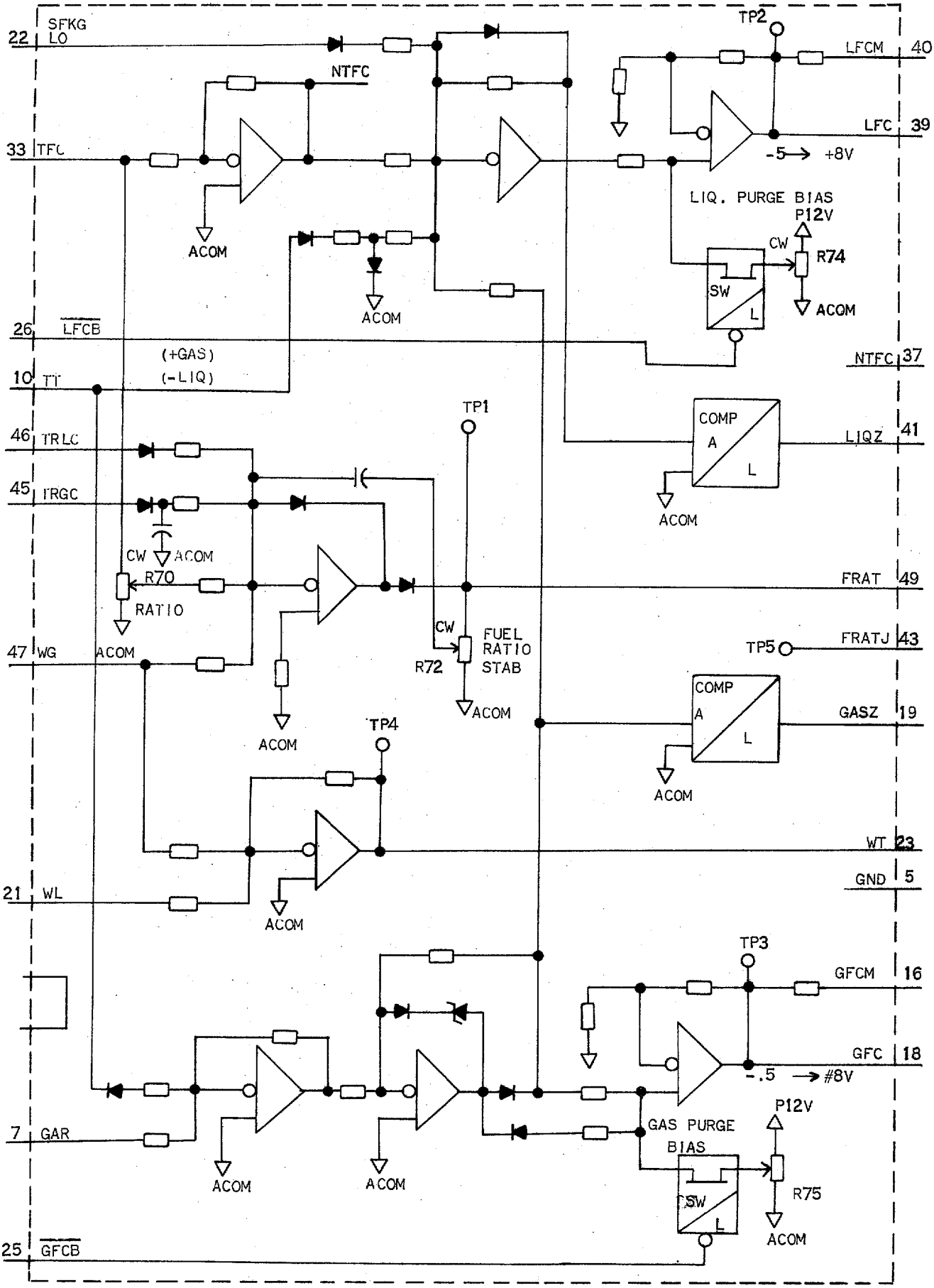


FIG. 6

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY GEK-40676, SH.#20

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS DRR	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM
REV. 2	ISSUED 6 3 74			FIRST MADE FOR GEK-40676		245A2260
REV. 3	MADE BY C. Andrews P Morrison			I.C. NO.		CONT. ON SH. FL. SH. NO. 10